POLITECNICO DI MILANO

Scuola di Ingegneria Industriale e dell'Informazione

Master Degree in Mechanical Engineering
Course of Advanced Dynamics of mechanical systems

REPORT ASSIGNMENT 2

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Introduction and overview of the problem

This assignment has the goal to analyze the mechanical behavior of a system through the use of a Finite Element Model: this report shows how to define its FE model, how to calculate the natural frequencies and mode shapes, how to compute different Frequency Response Functions, how to define the steady-state acceleration of the bike travelling on an irregular road, and lastly how to compute the static response of the structure subjected to the weight of the cyclist. In particular, the system taken into consideration is a bike frame made of aluminum ($E = 7 * 10^{10} \frac{N}{m^2}, \rho = 2700 \frac{kg}{m^3}$) whose beams have tubular cross section and has four lumped masses representing the wheels, the crankset and the handlebar. The beams of the bike can be categorized in three different classes depending on their properties and their lengths can be calculated starting from the coordinates of the nodes.

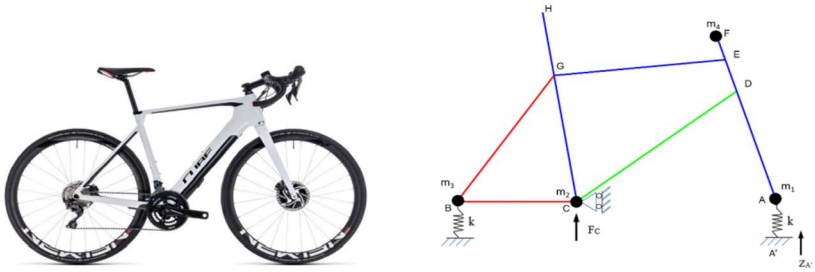
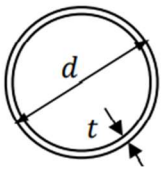


Fig. 1 Road bike: physical system and FE model

Point	x [m]	y [m]		Point	x [m]	y [m]
A	0.70	0		E	0.52	0.54
B	-0.42	0		F	0.49	0.63
C	0	0		G	-0.08	0.48
D	0.56	0.42		H	-0.12	0.72

Lumped masses	
m_1	1 kg
m_2	2 kg
m_3	2 kg
m_4	1.5 kg
Tyre stiffness (for 1 tyre)	
k	$4 \cdot 10^5 N/m$



Beam section	d [mm]	t [mm]
Red	20	1
Blue	32	1.5
Green	50	2.5

Fig. 2 Physical and geometrical properties

Definition of the FE model

To begin, it is necessary to define the positions of the nodes to subdivide the bike frame in finite elements: in the picture below it is showed the proposed discretization for the frequency range [0-200]Hz.

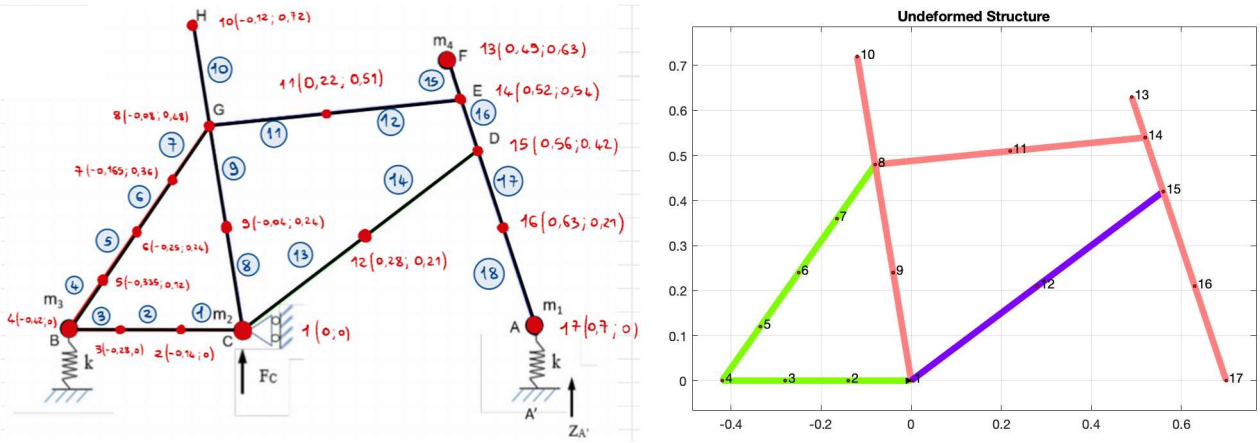


Fig. 3 Discretization of the structures

To proceed with the requests, firstly it is necessary to write a text file to assemble and draw the structure; the file is organized in the following way:

- List of nodes: number of the node - constraints (0 if the node is free, 1 if not) - x coordinate of the node - y coordinate of the node;
- List of elements (beams): number of the element - number of the input node - number of the output node - number of the assigned property;
- List of properties: number of the property - mass m - axial stiffness EA - bending stiffness EJ.

Then, it is needed to create the mass and stiffness matrices ([M] and [K] respectively). The first can be obtained by multiplying the diagonal matrices for the expansion matrix and its transposed and summing all contributes. The expansion matrix is full of zeros a part from the positions related to the d.o.f. of the nodes corresponding to the four masses, which can be individuated using the idb matrix. The result is then:

$$[M_i] = [E_{mi}]' * [M_{ih}] * [E_{mi}] \rightarrow M_{tot} = \sum_{i=1}^{4} [M_i]$$

On the other hand, [K] is obtained similarly multiplying the diagonal matrices for the expansion matrix and its transposed, but - since the springs are considered together - the sum is not needed:

$$[K] = [E_k]' * [K_h] * [E_k]$$

Natural frequencies and vibration modes

Once [M] and [K] have been assembled, the natural frequencies and mode shapes of the system can be computed: in particular, they are obtained by solving an eigenvalue problem.

[M] and [K] are square matrices with dimensions equal to the number of the d.o.f. (which are 51 corresponding to 17 nodes*3 d.o.f.) and can be partitioned in four sub-matrices: to evaluate natural frequencies and vibration modes it is sufficient to consider the free-free partitions (free d.o.f. inertia/stiffness seen by free d.o.f.). Since the free d.o.f. come before the constrained ones, the partitioned matrices can be obtained taking all the rows and columns from one to the number of free d.o.f. For this reason, the free-free matrices are usually in the left upper part of the initial matrices and have dimensions equal to the free d.o.f. (50, which is the initial number of d.o.f. minus the constrained one of node 1 due to the presence of the cart).

$$[M] = \begin{bmatrix} [M_{FF}][M_{FC}] \\ [M_{CF}][M_{CC}] \end{bmatrix}; \; [K] = \begin{bmatrix} [K_{FF}][K_{FC}] \\ [K_{CF}][K_{CC}] \end{bmatrix}$$

The natural frequencies and the mode shapes are both the outputs obtained by solving the following eigenvalues problem:

$$(\omega^2[I] - [M_{FF}]^{-1} * [K_{FF}])\underline{X} = 0$$

using the Matlab function *eig* which returns a diagonal of eigenvalues and a matrix whose columns correspond to the eigenvectors. The results can be sorted in ascending order so as to have correspondence between the two. The first four mode shapes and the corresponding natural frequencies can be visualized in the picture below:

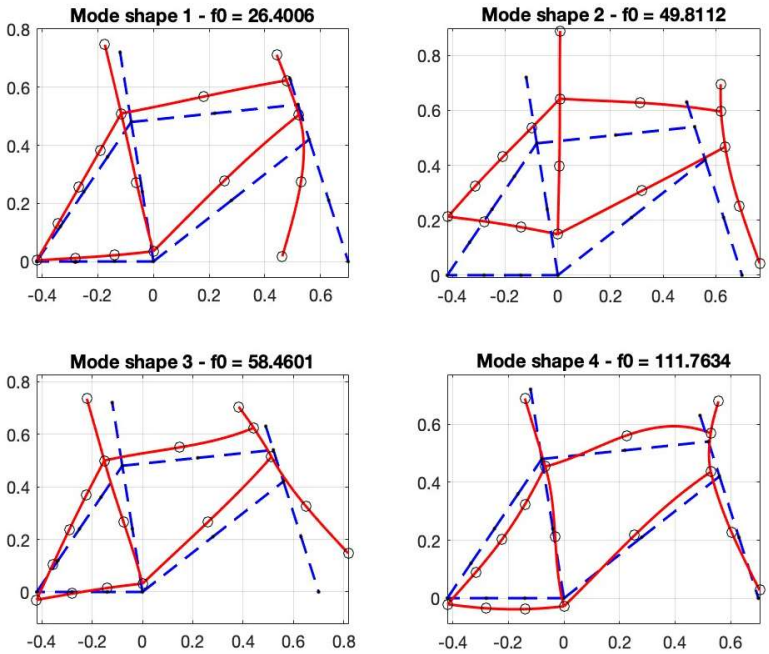


Fig. 4 First four mode shapes

Frequency response functions

The system presents a concentrated vertical force in the point C, identified by the second coordinate (vertical direction) of node 1 (idb(1, 2)), whose effect can be evaluated in terms of Frequency Response Functions that relate the input force to different outputs. They can be computed as shown:

$$(-\Omega^2[M_{FF}] + j\Omega[C_{FF}] + [K_{FF}])\underline{X}e^{j\Omega t} = \underline{F_0}e^{j\Omega t}$$

$$If\ [A] = (-\Omega^2[M_{FF}] + j\Omega[C_{FF}] + [K_{FF}]) \rightarrow [X] = A\backslash F_0$$

In general, it is possible to obtain as many FRFs as the number of the free d.o.f. of the system; but in this case some outputs are requested to be analyzed and the results are shown below. The frequency range considered is [0-200]Hz with a resolution of 0.1Hz, so $[X]$ has as rows as the free d.o.f. of the system and as columns as the length of the frequency interval.

Vertical displacement and acceleration of node F

These are taken out from [x], where the corresponding row coincide with the second coordinate (vertical direction) of node 13 (idb(13, 2)).

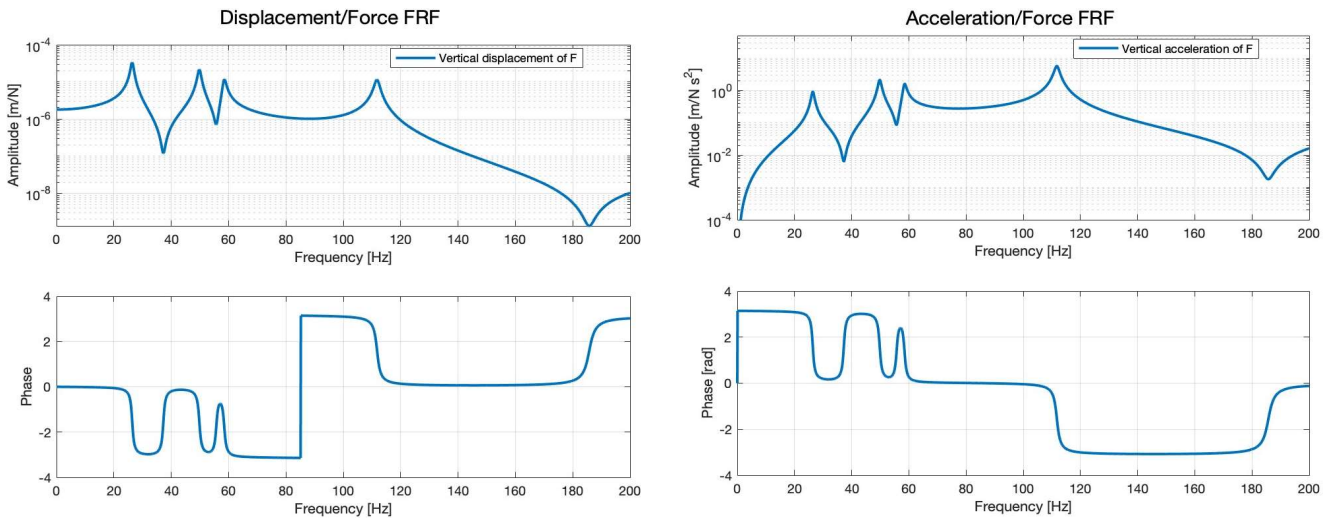


Fig. 5 Vertical displacement and acceleration of point F

Horizontal displacement and acceleration of node H

In this case, the corresponding row is identified by the first coordinate (horizontal direction) of node 10 (idb(10, 1)).

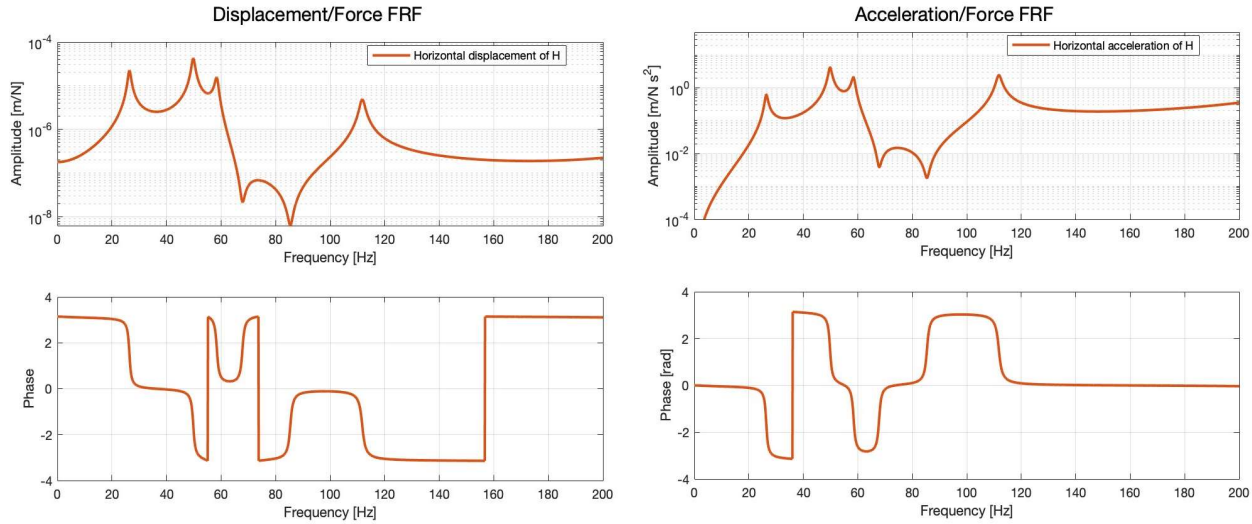


Fig. 6 Horizontal displacement and acceleration of point H

Shear and axial forces and bending moment of the midpoint of GE tube

In the FE model taken into consideration, the tube GE is divided in two elements (numbered 11 and 12) and its midpoint is the node 11. It is possible to evaluate the internal forces - shear and axial forces - and the bending moment in the node 11 both starting from the left (node 8) and the right (node 14) ends. For a modelled Bernoulli beam:

$$T = EJ(num_{el}) * (6 * d)$$

$$N = EA(num_{el}) * b$$

$$M = EJ(num_{el}) * (2 * c + 6 * d * \xi)$$

Firstly, the the 11th element on the left (delimited by nodes 8 and 11) has been considered. The results are presented in the figures below:

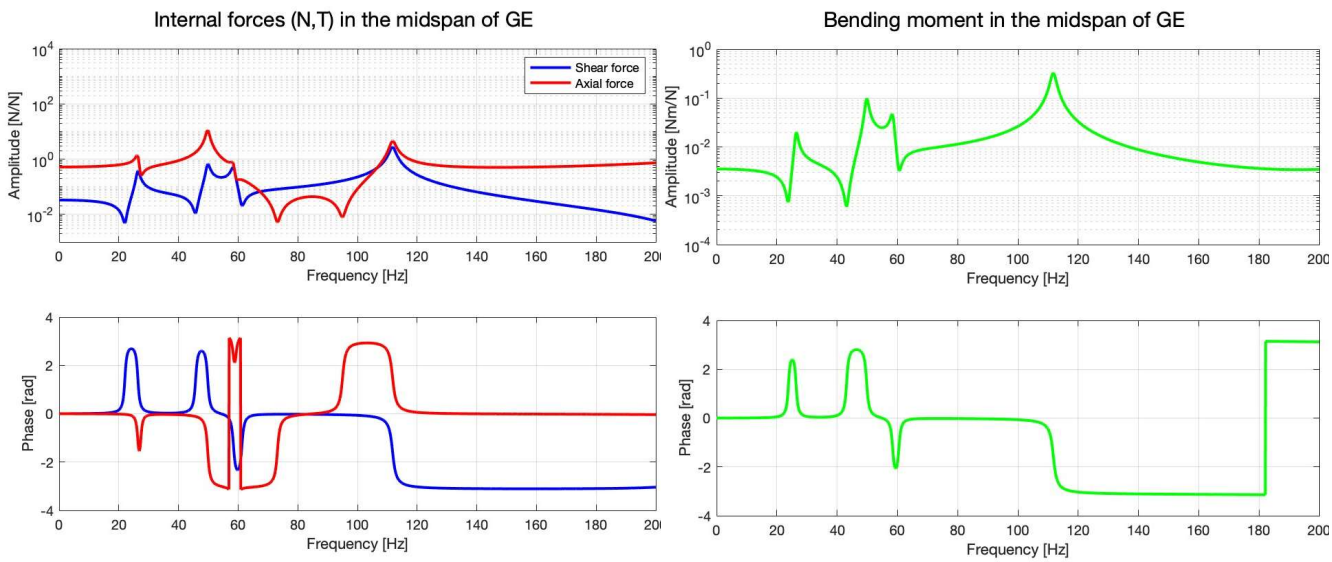


Fig. 7 Shear and axial forces and bending moment of the midpoint of tube GE

Then, it has been analyzed also the second case, the one considering the 12th right element (delimited by nodes 11 and 14). The results of the comparison between these last results and the previous ones are presented in the figure below:

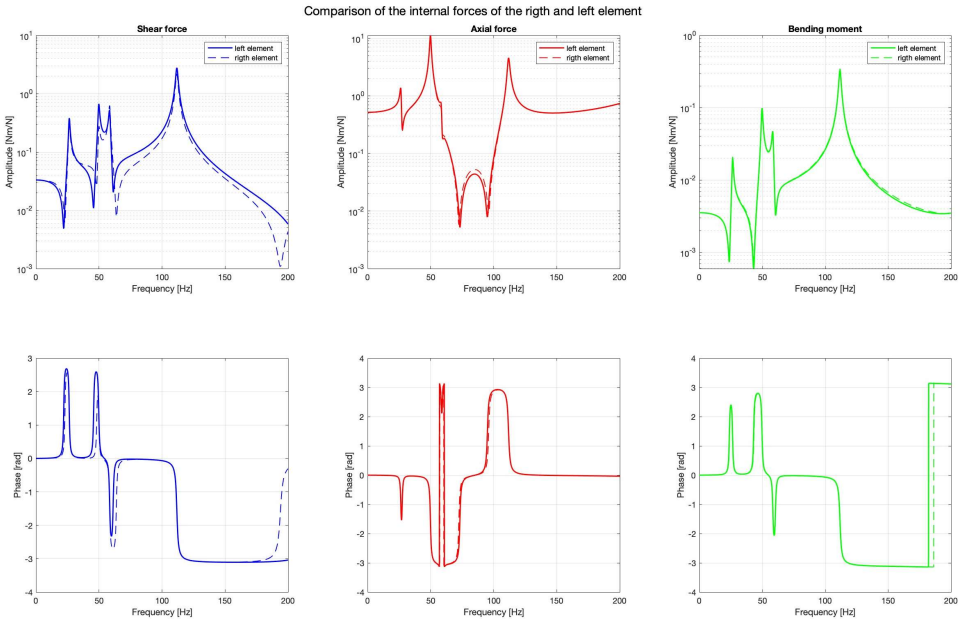


Fig. 8 Comparison of the two elements

It can be observed that, since the shear and axial forces are constant along an element, they depend on the choice of the element. On the other hand, the bending moment is linear, so it assumes the same value in the 11th node, which is contemporaneously the final point for element 11 and the starting one for element 12.

Constraint force in node C

The node C is connected to the ground through a cart, which prevents the possibility of that point to move horizontally. For this reason, removing the constraint, it is possible to evaluate the horizontal constraint force in point C (node 1):

$$\underline{R} = (-\Omega^2[M_{CF}] + j\Omega[C_{CF}] + [K_{CF}]) * \underline{X}$$

The magnitude and the phase of the FRF relating the vertical force imposed in C and the constraint force are shown below:

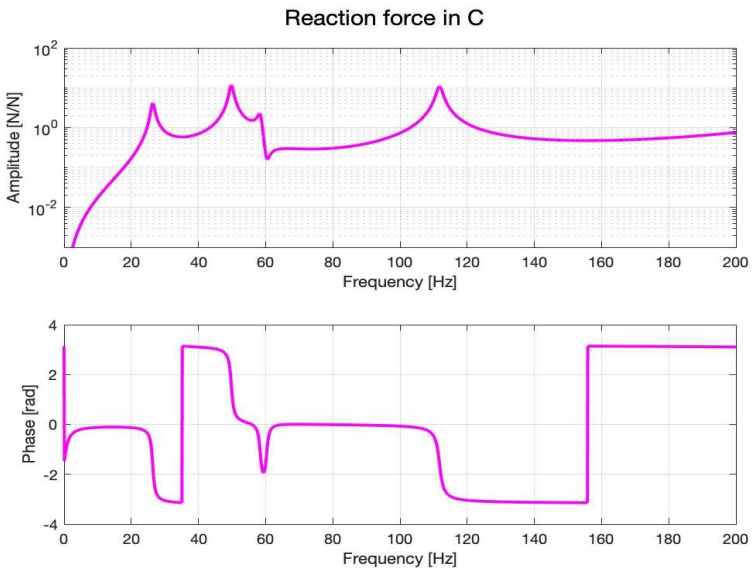


Fig. 9 Constraint forces of point C

Application of the modal superposition approach

The modal superposition approach consists of introducing a new set of coordinates - the modal ones - in order to decouple the EOM obtaining diagonal matrices, to discretize the system and get a simple and effective method of managing forced vibrations. To do this, after defining a new matrix collecting the modal vectors, it is required to evaluate the new mass, stiffness and damping matrices by multiplying the original ones for the modal matrix and its transposed. The resulting matrices are particularly significant because they are diagonal - the modal damping matrix is diagonal in case of light structural damping - since they are linear combinations of diagonal matrices. In the case taken into consideration, it is requested to consider the structure's first two modes, so:

$$[\phi] = [\underline{X^{(1)}}\ \underline{X^{(2)}}]$$

$$[M_q] = [\phi]^T[M][\phi];\ [K_q] = [\phi]^T[K][\phi];\ [C_q] = [\phi]^T[C][\phi];\ [Q_q] = [\phi]^T\underline{F}$$

Frequency Response Functions

At this point, it is possible to evaluate the new Frequency Response Functions which are expressed in principal coordinates (FRFs in modal superposition approach). These are evaluated starting from:

$$(-\Omega^2[M_q] + j\Omega[C_q] + [K_q])\underline{X_q}e^{j\Omega t} = \underline{F_q}e^{j\Omega t} \rightarrow \underline{X_q} = (-\Omega^2[M_q] + j\Omega[C_q] + [K_q])\backslash\underline{F_q}$$

The input taken into consideration is still the vertical force concentrated in point C, whereas the outputs are the ones presented below. The resulting diagram are presented as superimposed to the previous ones.

Vertical displacement and acceleration of node F

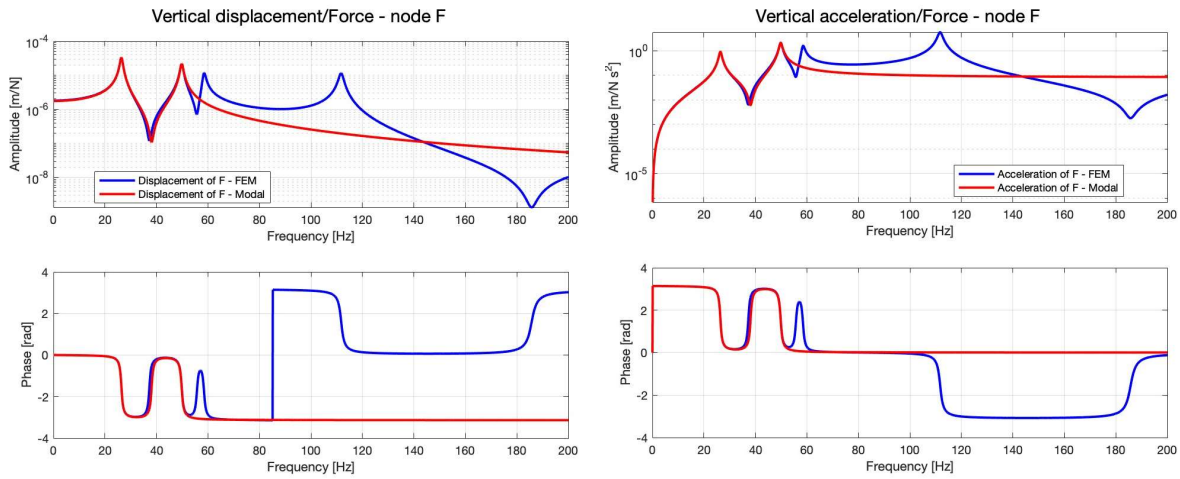


Fig. 10 Vertical displacement and acceleration of point F - Modal approach

Horizontal displacement and acceleration of node H

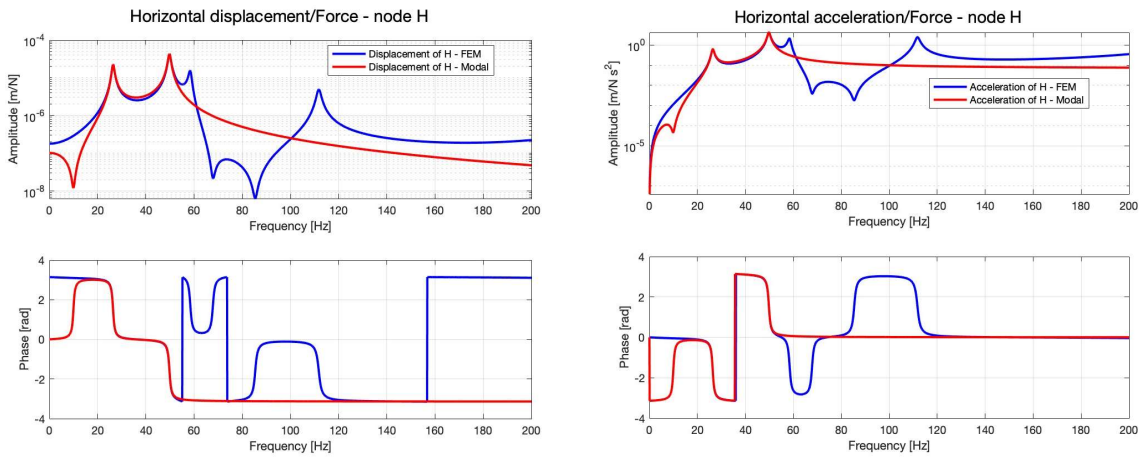


Fig. 11 Horizontal displacement and acceleration of point H - Modal approach

Results

It can be observed that in all the cases taken into consideration - both for displacement and acceleration and for nodes F and H - the first two modes are excited. The modal approach fits adequately these two mode shapes, but it is worth noticing that it does not at the beginning of the horizontal displacement and acceleration maybe because it is influenced by the other mode shapes that are neglected in the application of the modal superposition approach.

Motion imposed at the constraint

The point A (corresponding to the 17th node) is connected by a spring to the point A' (19th node), which imposes a displacement z_0 in the vertical direction, so the point A (idb(17, 2)) is subjected to z_0. This can be considered a new input - varying in the frequency range [0-200]Hz with a resolution of 0.1Hz - of the system and plotting the Frequency Response Functions it is possible to discuss how it influences the vertical acceleration of point F (node 14). Similarly to before, the displacement can be evaluated as following:

$$-(-\Omega^2[M_{FC}] + j\Omega[C_{FC}] + [K_{FC}])\underline{X_{imp}}e^{j\Omega t} = \underline{F_0}e^{j\Omega t}$$

$$If\ [A] = (-\Omega^2[M_{FF}] + j\Omega[C_{FF}] + [K_{FF}]) \rightarrow [X_{imp}] = A\backslash F_0$$

and the acceleration:

$$[\ddot{X}_{imp}] = [X_{imp}] * -\Omega^2$$

The results are shown below:

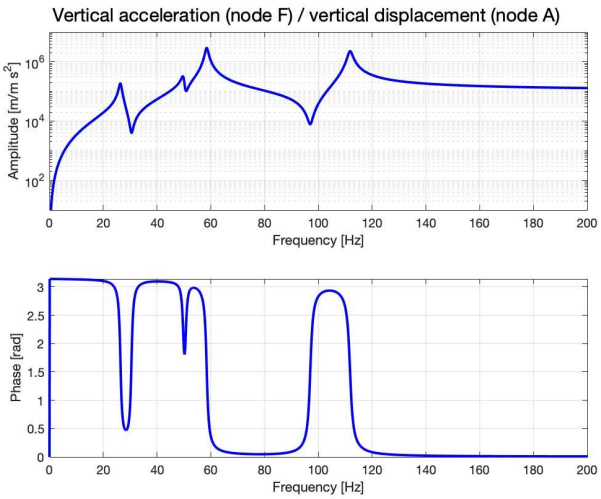


Fig. 12 FRF due to the motion imposed at the point A

Bike travelling on the road and superposition principle

Consider the bike taken into consideration travelling on a road with irregularities described by two wavelengths with a speed of 12 m/s. Relying on the superposition principle, it is possible to evaluate the steady-state vertical acceleration of point H by assessing the contribute of each singular wavelength and then summing up both the results. Let's discuss the procedure followed referring to the first curve; then, the process can be iterated for the second.

Knowing the speed and the wavelengths λ of the curves, it is possible to calculate the frequency in Hertz and, so, the angular frequency in radians:

$$\lambda_1 = \frac{v}{f_1} \rightarrow f_1 = \frac{v}{\lambda_1} \rightarrow \omega_1 = 2\pi f_1$$

Considering that at the beginning (time t=0) the center of the front-wheel is located above a positive maximum of the irregularity profile, the motion imposed at the front-wheel corresponding to the node A' (downside the spring) can be expressed as cosinusoidal functions:

$$z_{A1}(t) = A_1 * \cos(\omega_1 t + \varphi_{A1}), where\ \varphi_{A1} = 0$$

On the other side, the motion imposed at the back-wheel depends on the time delay between the two wheels: the distance between the two wheels equals the distance between nodes 4 and 17, so:

$$t_{delay} = \frac{distance}{speed}$$

Then, because point B has some delay with respect to point A, the motion imposed in B is:

$$z_{B1}(t) = A_1 * \cos(\omega_1 t + \varphi_{B1})$$

φ_{B1} can be evaluated considering that, due to the time delay between the two wheels, the position of A at a time equal to zero is equal to the one of B evaluated in an instant equal to the delay:

$$z_{A1}(0) = A_1 * \cos(\omega_1 * 0) = z_{B1}(t_{delay}) = A_1 * \cos(\omega_1 * t_{delay} + \varphi_{B1}) \rightarrow \varphi_{B1} = -2 * \pi * f_1 * t_{delay}$$

Since two springs are connected to the points A and B, the presence of these imposed motions induces two forces:

$$F_A = k * z_{A1} = k * \widehat{A_1} * e^{j\omega_1 t}; \qquad F_B = k * z_{B1} = k * \hat{A} * e^{j\omega_1 t} * e^{j\varphi_{B1}}$$

If $\hat{A}$ is the amplitude of the first curve and F_1 is a vector collecting all the inputs, the only non-zero elements are located in the positions that identify the vertical d.o.f. of the two wheels (idb(4, 2) for wheel B and idb(17, 2) for the wheel A. Subsequently, the displacement and the acceleration of each point of the frame can be evaluated:

$$(-\omega_1^2[M_{FF}] + j\omega_1[C_{FF}] + [K_{FF}]) * \underline{x_1} = F_1$$

$$\underline{x_1} = (-\omega_1^2[M_{FF}] + j\omega_1[C_{FF}] + [K_{FF}])\backslash F_1 \rightarrow \underline{\ddot{x_1}} = (-\omega_1^2) * \underline{x_1}$$

The requested vertical acceleration of point H - equal to node 10 - is assessed by finding the corresponding contribute in the vector $\underline{\ddot{x_1}}$, which is identified by idb(10, 2):

$$\underline{\ddot{x_{1H}}} = \left|\underline{\ddot{x_{1H}}}\right| * \cos(\omega_1 t + \varphi_{H1})$$

where $\left|\underline{\ddot{x_{1H}}}\right|$ is the amplitude of the acceleration vector and φ_{H1} the phase. The resulting acceleration of point H evaluated in time for the first input force is shown below:

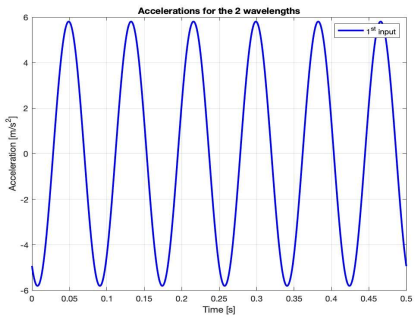


Fig. 13 Acceleration of point H due to the first wave

Then, it is possible to replicate the procedure for the second wave finding the second input force F_2 and the corresponding resulting acceleration of point H evaluated in time, which is the sequent:

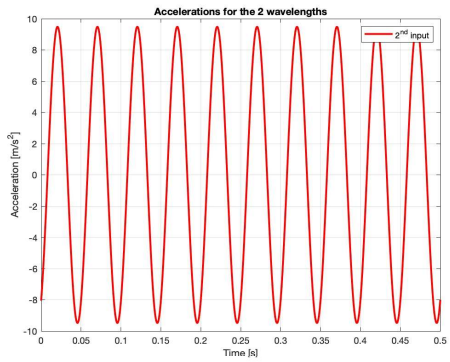


Fig. 14 Acceleration of point H due to the second wave curve

Then, by applying the superposition principle (so summing up the two contributes), it is possible to plot the overall time history of the steady-state vertical acceleration of point H. The result is the following:

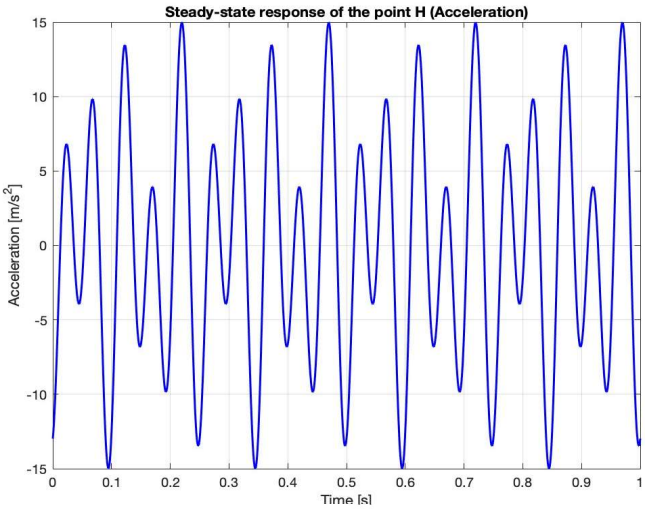


Fig. 15 Acceleration of point H by applying the superposition principle

Static response to the weight force

It is possible to consider the bike subjected to a weight force due to the presence of a cyclist, which can be represented by two vertical forces: the major applied in the saddle (point H, node 10, having vertical direction, so identified by idb(10, 2)), the minor in the handlebar (point F, node 13, having vertical direction, so identified by idb(13, 2)). In this condition, the bike is deformed and shows a static response.
If W_0 is a column vector with a number of rows equal to the d.o.f. of the structure and having non-zero elements only in the positions identified before (idb(10, 2) and idb(13, 2)), the displacement can be evaluated as following:

$$[K_{FF}] * x_F = W_0 \rightarrow x_F = [K_{FF}] \backslash W_0$$

In this way, the deformed shape of the bike frame can be compared to the undeformed configuration, as shown below:

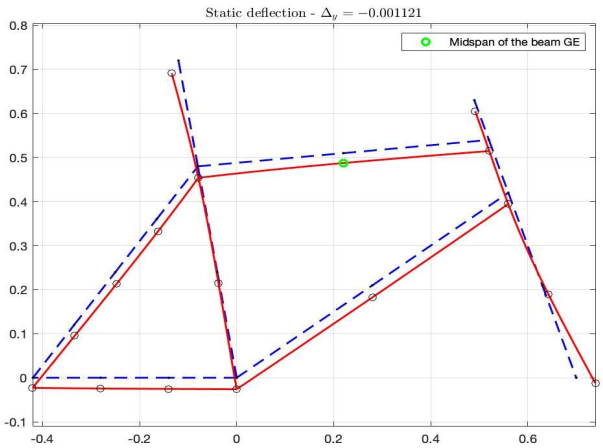


Fig. 16 Static deflection

Then, considering the midpoint of the tube GE (equivalent to node 11), it is possible to assess its displacement taking from the vector x_F only the elements corresponding to the node taken into consideration and so identified by idb(11, 1) for the horizontal one and idb(11, 2) for the vertical:

$$x_{F,vert} = x_F\big(idb(11,2)\big);\ x_{F,hor} = x_F\big(idb(11,1)\big)$$

Analyzing the results, it can be stated that the vertical displacement of this point is approximately equal to $\Delta_y = -0.001121$.